

SQL Essentials

Custom knowledge-base material for the AI & Data Science RAG Chatbot project.

1. What is SQL?

SQL stands for Structured Query Language. It is used to store, retrieve, update, and analyze data in relational databases. Tables contain rows and columns, and relationships are commonly represented using keys.

2. SELECT and Filtering

SELECT chooses columns. FROM identifies a table. WHERE filters rows before grouping. ORDER BY sorts results. DISTINCT removes duplicate result rows. LIMIT restricts the number of returned rows in supported databases.

3. Aggregate Functions

COUNT counts rows or non-null values depending on usage. SUM, AVG, MIN, and MAX summarize numeric data. GROUP BY creates groups, while HAVING filters groups after aggregation.

4. Joins

INNER JOIN returns matching rows from both tables. LEFT JOIN keeps all rows from the left table and matching rows from the right. RIGHT JOIN keeps all rows from the right. FULL OUTER JOIN returns rows from either side where supported.

5. Subqueries and CTEs

A subquery is a query inside another query. A Common Table Expression, written with WITH, creates a named temporary result that improves readability and can be reused within a statement.

6. Window Functions

Window functions calculate values across related rows without collapsing them. ROW_NUMBER assigns sequential numbers, RANK handles ties with gaps, and LAG accesses a previous row. They commonly use OVER with PARTITION BY and ORDER BY.

7. Keys and Normalization

A primary key uniquely identifies a row. A foreign key references a key in another table. Normalization reduces duplication and update anomalies by organizing data into related tables.

8. Best Practices

Select only needed columns, use clear aliases, format queries consistently, validate joins to avoid accidental row multiplication, and use parameterized queries in applications to reduce SQL injection risk.